

A PROJECT TITLE
ON
Heart Disease Prediction Using Logistic Regression, SVM, and K-Nearest Neighbors
project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY
in
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DEPARTMENT OF CSE

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1. ABSTRACT:

Heart disease is one of the major health concerns worldwide, and its early prediction can support timely diagnosis and treatment. This project focuses on developing a machine learning-based heart disease prediction system using Logistic Regression, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). The study uses a heart disease dataset containing 1,025 patient records with 13 clinical and demographic features, including age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, resting ECG results, maximum heart rate, exercise-induced angina, ST depression, slope, number of major vessels, and thalassemia. The target variable indicates whether heart disease is present or absent. The dataset is analyzed and preprocessed before model training, including handling duplicate records, separating input features and the target variable, splitting the data into training and testing sets, and applying feature scaling where required. The three classification algorithms are trained and evaluated using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC. Their results are compared to identify the most effective model for predicting heart disease using the given dataset. The project demonstrates the practical application of machine learning in healthcare by analyzing patient-related attributes and predicting the likelihood of heart disease. The developed system can serve as a basic decision-support tool for early risk identification, while final medical diagnosis should always be performed by qualified healthcare professionals. We are using the Heart Disease dataset provided by Kaggle.

2. INTRODUCTION:

Heart disease is among the most serious health concerns globally and is responsible for millions of deaths each year. Early identification of individuals at risk plays a crucial role in reducing mortality rates and improving treatment outcomes. However, diagnosing heart disease accurately often requires experienced physicians, multiple medical tests, and considerable time. Machine Learning has emerged as a powerful technology that can analyze large amounts of healthcare data and identify hidden patterns that assist in disease prediction. This project focuses on predicting heart disease using three widely used supervised machine learning algorithms: Logistic Regression, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). These algorithms are trained using historical patient data containing clinical attributes such as age, blood pressure, cholesterol, chest pain type, and maximum heart rate. By learning relationships between these

features and the target variable, the models can predict whether a patient is likely to have heart disease.

The project also compares the performance of these algorithms using evaluation metrics including Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Such a comparative analysis helps determine the most effective algorithm for this classification problem. The developed prediction system can serve as a decision-support tool for healthcare professionals by providing quick and reliable predictions. Ultimately, this project demonstrates how machine learning techniques can improve medical diagnosis, reduce healthcare costs, and contribute to better patient care through early disease detection.

3. SOFTWARE REQUIREMENTS:

Developing a Heart Disease Prediction system requires both suitable software tools and adequate hardware resources to ensure efficient model training and evaluation. The software stack provides the environment for data preprocessing, model training, and evaluation, while the hardware ensures smooth execution of the machine learning workflow. Together, they form the backbone of the project, enabling accurate and efficient prediction of heart disease.

Requirement Type	Specification
Software	Operating System: Windows 10/11, Linux, or macOS
	Programming Language: Python 3.10 or later
	IDE: Jupyter Notebook / Google Colab
	NumPy, Pandas (for data handling)
	Matplotlib, Seaborn (for visualization)
	Scikit-learn (for model implementation)
Dataset	Heart Disease Dataset (CSV format, sourced from Kaggle)
	Algorithms: Logistic Regression, Support Vector Machine (SVM)
	K-Nearest Neighbors (KNN)


Signature of the Guide


Course Instructor